

Title: Ouroboros Versus EWT+QCD: A Compact, Falsifiable Comparison with Emphasis on Dark-Matter Phenomenology

Abstract The Ouroboros Lagrangian is a compact classical field theory that, when calibrated to the electron, claims to reproduce a range of particle physics observables while using only three free parameters. It predicts lepton masses, the elementary charge, the proton radius, and a novel dark-matter soliton — the neutral chaoiton — with $m\chi = 0.460$ MeV and a light bosonic mediator $mJ = 0.618$ MeV. Angular-momentum selection rules produce a vanishing monopole coupling and a strongly dipole-suppressed nucleon cross section, which the authors argue reconciles the model with direct-detection and astrophysical bounds. This paper presents a transparent, reproducible comparison to the Standard Model (EWT+QCD), identifies the strongest independent theoretical and experimental checks, and proposes a prioritized falsification roadmap centered on NEGF vertex calculations, sub-MeV searches, and a J-field sensor test for a six-peak annual modulation tied to Gaia stellar streams. Canonical numerical profiles and scripts are archived with DOI 10.5281/zenodo.20612600.

1. **Introduction** The Standard Model (EWT+QCD) is an empirically successful framework for particle interactions but relies on many empirically fixed parameters and leaves cosmological dark matter unexplained. The Ouroboros program proposes a different approach: a superrenormalizable classical field theory with three core parameters calibrated to the electron that reportedly yields quantitative predictions for multiple observables and a concrete dark-matter candidate. This paper compares the empirical reach, formal structure, and testability of Ouroboros against EWT+QCD, with a focus on the model's most distinctive, falsifiable claims. We separate established, reproducible results from conjectural extensions and provide a prioritized list of theoretical and experimental checks required to validate or falsify the central claims.
2. **Technical overview of Ouroboros**
2.1 Lagrangian and fields The model uses an electromagnetic four-potential A_μ and a massive vector J-field J_μ with a quartic self-interaction. The neutral sector admits an exact scaling symmetry that fixes the J-field mass scale independently of the quartic amplitude. The Euler–Lagrange equations couple A_μ and J_μ and permit both localized soliton solutions and delocalized wave modes.

2.2 Calibration and parameter economy Three parameters (commonly denoted g , λ , and ω in the Ouroboros literature) are fixed by matching the electron energy scale and related observables. With that calibration the authors report derived values for other observables (muon, tau, elementary charge, proton radius) without introducing new free parameters.

2.3 Solitons and excitations The neutral chaoiton is an $l = 1$ (p-wave) localized soliton solution of the neutral-sector boundary-value problem. The same Lagrangian admits delocalized J-field wave modes; the model therefore naturally supports a two-component dark-matter phenomenology (localized chaoitons + J-field waves).

2.4 Formal claims and artifacts The authors assert superrenormalizability and provide machine-verified Lean 4 proofs for several mathematical ingredients (linking number, mountain-pass existence, power counting). Independent numerical reproduction of the canonical $\beta(r)$ profile has been reported.

3. Empirical comparison: decision-relevant attributes 3.1 Compact comparison summary

Free parameters

- Ouroboros (claims): 3 (g, λ, ω)
- EWT + QCD (Standard Model): 19+ (Yukawas, mixing angles, Higgs mass, couplings)

Lepton masses

- Ouroboros (claims): Predicts muon & tau from harmonics (reported gaps: 0.8% and 6.5%)
- EWT + QCD: Input parameters (not predicted)

Elementary charge

- Ouroboros (claims): Predicted within 0.6%
- EWT + QCD: Input parameter

Proton radius

- Ouroboros (claims): Predicted ≈ 0.84 fm (from R_{phys})
- EWT + QCD: Determined by QCD / lattice calculations

Dark-matter candidate

- Ouroboros (claims): Neutral chaoiton $m_\chi = 0.460$ MeV; $m_J = 0.618$ MeV
- EWT + QCD: No soliton candidate; DM external to SM

Direct-detection compatibility

- Ouroboros (claims): Dipole suppression $\rightarrow \sigma p \sim 6.7 \times 10^{-42} \text{ cm}^2$ (claimed)
- EWT + QCD: N/A (model-dependent)

Formal verification

- Ouroboros (claims): Lean 4 proofs claimed; independent numerical reproduction reported
- EWT + QCD: Extensive community validation for perturbative QFT; lattice QCD for nonperturbative results

Primary empirical tests

- Ouroboros (claims): NEGF vertex check; sub-MeV searches; six-peak Gaia modulation
- EWT + QCD: Precision collider tests; neutrino, cosmology constraints

3.2 Interpretation Ouroboros claims a compact parameter set with several quantitative outputs; the Standard Model trades compactness for a broad, experimentally validated dynamical framework. The comparison highlights that Ouroboros makes bold, falsifiable predictions that can be tested independently; it does not claim to immediately supplant the SM but to offer an alternative with distinct, testable consequences.

4. Dark-matter phenomenology and falsifiable predictions
- 4.1 Neutral chaoiton summary • Soliton type: $l = 1$ localized soliton (neutral chaoiton). • Masses: $m_\chi = 0.460$ MeV, $m_J = 0.618$ MeV. • Selection rule: Monopole electromagnetic coupling vanishes by angular-momentum orthogonality; leading coupling is dipole. • Dipole suppression: Small- q expansion of the dipole form factor yields strong suppression at galactic momentum transfers; combined with nuclear-scale vertex suppression the reported chaoiton–proton cross section is $\sigma_p \approx 6.7 \times 10^{-42} \text{ cm}^2$. • Self-interaction: Unsuppressed Born estimate is large, but dipole form factors are claimed to suppress effective σ/m to $\lesssim 10^{-6} \text{ cm}^2/\text{g}$, compatible with Bullet Cluster bounds.

4.2 Two-component phenomenology and six-peak modulation The model allows a mixture of localized chaoitons and delocalized J-field waves; the local chaoiton/wave ratio depends on formation history. A distinctive, falsifiable signature is a six-peak annual modulation in local J-field flux tied to Gaia stellar streams (peaks in June–December at stream-dependent dates). Detection of this pattern with a sufficiently sensitive J-field sensor would strongly support the model; a null result from a sensor with the claimed sensitivity would falsify the chaoiton as the dominant local DM component.

4.3 Cosmology and relic abundance (sketch) Because the chaoiton couples extremely weakly (dipole suppressed), freeze-in production is the natural mechanism for relic abundance, avoiding thermal equilibrium and large contributions to N_{eff} . A full momentum-dependent Boltzmann integration is required to confirm this quantitatively.

4.4 Key theoretical checks required

1. NEGF vertex calculation to confirm that non-linear dressing does not reintroduce an unsuppressed monopole component and to compute the exact dipole form factor including vertex corrections.
2. Chaoiton–chaoiton scattering vertex (dipole–dipole angular integrals) to verify the claimed suppression of self-interaction at cluster velocities.
3. Cosmological integration of the coupled χ –J Boltzmann system including momentum dependence and freeze-in channels to verify BBN/CMB consistency and structure formation constraints.
4. Reproducibility, data, and experimental roadmap 5.1 Minimum reproducibility deliverables • Canonical $\beta(r)$ profile CSV with header and units. • Dipole form factor $F(q)$ CSV and plotting script. • Exact environment (requirements.txt / environment.yml) and example commands with seeds. • Lean 4 proof artifacts with build instructions and short human-readable summaries. • NEGF code or algorithmic description for vertex corrections.

5.2 Prioritized experimental roadmap Priority A (theory): NEGF vertex and chaoiton–chaoiton vertex calculations. These determine whether the dipole suppression claims survive many-body dressing. Priority B (data reanalysis): Reanalysis of existing time-series electromagnetic data for the six-peak template; cross-correlate with Gaia stream kinematics. Priority C (instrument): Develop a prototype J-field sensor (quantum-dot array + matched-filter processing) and perform a year-long deployment to search for the six-peak modulation. Priority D (direct search): Design a dedicated sub-MeV dark-matter experiment sensitive to the predicted masses and the model's interaction signature.

5.3 Practical steps for community verification • Publish canonical data and scripts with exact environment and seeds. • Release Lean 4 artifacts and short proof summaries to allow independent verification. • Commission independent NEGF and many-body vertex analyses. • If theoretical checks pass, prioritize instrument development and data reanalysis.

6. Limitations, open questions, and risk assessment 6.1 Main open theoretical issues • The NEGF vertex calculation is essential; if vertex dressing reintroduces an unsuppressed monopole component the model's direct-detection compatibility would be compromised. • The chaoiton–chaoiton scattering amplitude requires a careful multipole angular integration; the simple factorization used in preliminary estimates may be insufficient. • Cosmological freeze-in must be computed with the momentum-dependent dipole form factor to confirm BBN/CMB constraints and structure formation viability.

6.2 Empirical risks • The tau mass prediction gap (reported $\approx 6.5\%$) is larger than the muon gap and may indicate missing physics or calibration sensitivity. • The six-peak modulation is a strong, specific signature but requires a sensor concept and sensitivity that are not yet demonstrated. Null results from sufficiently sensitive searches would falsify the local chaoiton dominance.

7. Conclusion and recommended next steps Ouroboros presents a compact, falsifiable alternative that makes concrete numerical predictions and supplies formal artifacts (Lean 4 proofs, numerical profiles). The most decisive near-term work is theoretical: NEGF and vertex analyses that confirm or refute the dipole suppression mechanism. If those checks pass, the experimental program (J-field sensor, sub-MeV searches, Gaia-template reanalysis) becomes urgent and compelling. If they fail, the model still contributes useful ideas (soliton dark matter, angular-momentum selection rules) that can inform broader model building.

Appendix A — Notation table • $\mathcal{L}_J$ — Ouroboros Lagrangian density; units: energy⁴ in natural units. • A_μ — Electromagnetic four-potential; units: energy. • J_μ — J-field four-potential (massive vector mediator); units: energy. • g — Quartic self-coupling (dimensionless in code units; set to 1 for electron calibration). • λ — Scaling parameter controlling neutral-sector ODE; dimensionless. • ω — Harmonic index used in lepton mass sequence; dimensionless. • $\beta(r)$ — Radial profile of J^0 for the $l = 1$ soliton; units: dimensionless. • m_χ — Chaoiton mass; units: MeV. • m_J — J-field mediator mass; units: MeV. • C — Effective coupling constant used in Born cross sections; units: MeV·fm. • $F_{\text{dipole}}(q)$ — Dipole form factor; dimensionless. • σ_p — Chaoiton–proton cross section; units: cm². • R_{dipole} — Dipole radius defined by $\int \beta r^3 / \int \beta r^2$; units: fm.

Appendix B — Reproducibility appendix (example commands and environment) Recommended environment • Python 3.10.x • numpy, scipy, pandas, matplotlib (versions pinned in requirements.txt)

Example requirements.txt (suggested) `python==3.10.* numpy==1.24.* scipy==1.10.* pandas==2.2.* matplotlib==3.7.*`

Example commands • Reproduce canonical $\beta(r)$: `python scripts/compute_beta.py --lambda -0.264697 --seed 42 --out v11_canonical_beta_profile.csv` • Compute dipole form factor: `python scripts/compute_Fq.py --beta v11_canonical_beta_profile.csv --qmax 100 --out Fq_dipole.csv` • Plot modulation: `python scripts/plot_modulation.py --phi Phi_time_series.csv --out Phi_annual_modulation.png`

Data files (recommended headers) • `v11_canonical_beta_profile.csv` — columns: `r_fm [fm]`; `beta [dimensionless]`; `dbeta_dr [dimensionless/fm]`. • `Fq_dipole.csv` — columns: `q_MeV [MeV]`; `Fq [dimensionless]`. • `Phi_time_series.csv` — columns: `date [ISO]`; `Phi [cm-2 s-1]`.

Appendix C — Suggested figure captions (publication ready) Figure 1. Canonical $\beta(r)$ profile for the neutral chaoiton. Horizontal axis: radius r in fm; vertical axis: $\beta(r)$ (dimensionless). The inset shows the small- r linear fit used to extract B_0 . Data source: v11_canonical_beta_profile.csv.

Figure 2. Dipole form factor $F_{\text{dipole}}(q)$ vs momentum transfer q . Horizontal axis: q in MeV; vertical axis: $|F_{\text{dipole}}(q)|$ (dimensionless). The small- q analytic expansion $|F| \approx (q/3) R_{\text{dipole}}$ is overlaid as a dashed line. Data source: Fq_dipole.csv.

Figure 3. Predicted six-peak annual modulation curve $\Phi(t)$ from Gaia stream templates. Horizontal axis: calendar date; vertical axis: flux Φ in $\text{cm}^{-2} \text{s}^{-1}$. Peaks correspond to stream kinematics listed in Table 2 of the manuscript.